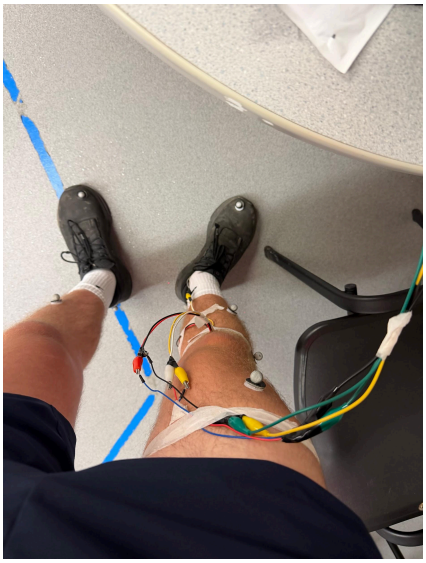
Prototyping 4

Michael Phelan, Concordia Lutheran High School | Dr. T. Jenkyn, Western University |

For this prototyping session, I went to the Wolf Orthopaedic Biomechanics Laboratory and tested my device. Dr. Jenkyn and Ms. Costello (a grad student) were both kind enough to help me.

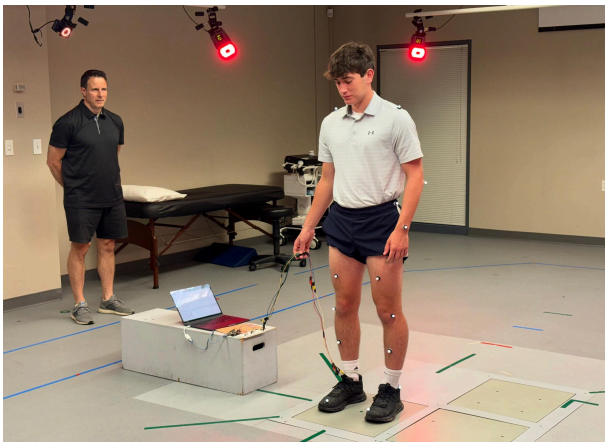
We had 19 tests in total, with each test having 3 trials each. The point of this session was to find out how accurate my device was, and to figure out what is wrong with it if something is off. My device can only estimate force from how much two sensors are being squashed. The force plate measures force directly. So for every trial I recorded what my device said and Ms. Costello recorded what the force plate said, and wherever the two disagree, the force plate is right. For each trial, Ms. Costello and I would line up our times. We both hit “go” at roughly the same time, and recorded for 5 seconds.

For the percentage tests, I would press down on the forceplate with my device until Ms. Costello had told me I had hit the target (some percent of my bodyweight in newtons) before collecting the data. The moving tests (Walking slow/Walking normal/Speed Walking/Running) had no target; I just moved across the plate. Featherweight and Flat foot press had no target either, as we were measuring how accurate terms used by doctors were.



For all the tests, I had the insole connected to my computer via alligator clips and wires, which were taped down on me. Reflective markers were also put on me for gait analysis.

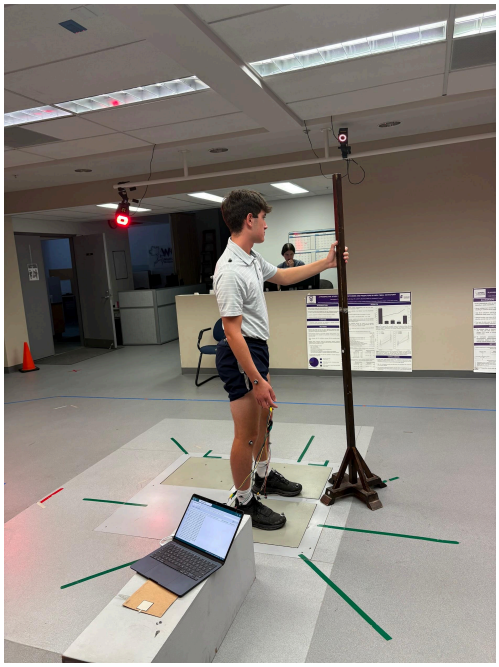
Here are all the tests done (3 trials each):



Stand on both feet full weight bearing: I stood normally with both feet on the force plate. This is the only test where both feet were on the plate, which means the plate measured my entire body weight rather than just my test foot. To compare it against my device I had to assume my weight was split evenly between my two feet.

Stand on one foot full weight bearing: I stood on just my right foot (the one with the device) on the force plate, with my left foot lifted. This put essentially all of my body weight on the sensor.

30 percent: I put my right foot with the device in it on the force plate. My left foot was to the side, not on the plate. I would press down with my right foot until Ms. Costello told me I was standing on 30 percent of my body weight, then we would each record 5 seconds of our values.



*Note: for the three trials above, we had to redo them, as the first time they were done, the ball sensor was accidentally disconnected

50 percent: Same as 30% test, except I pressed down until Ms. Costello told me I was standing on 50 percent of my bodyweight

85 percent: Same as 30% test, except I pressed down until Ms. Costello told me I was standing on 85 percent of my bodyweight

Featherweight: This is a common term that doctors use to prescribe under 10% weight bearing. This was the lightest touch I could manage while still having my foot on the force plate

Flat foot press: This is a term that doctors use to prescribe ~ 25% weight bearing. For this, I just rested my full foot against the force plate

Walking slow/Walking normal/Speed Walking/Running: I moved across the force plate at four different speeds. The plate only captures one step, so each trial recorded a single footfall of my right foot landing on it. However, my device recorded the entire 5 seconds, which included multiple strides. There were three trials each, so 12 trials total for these “moving” tests.



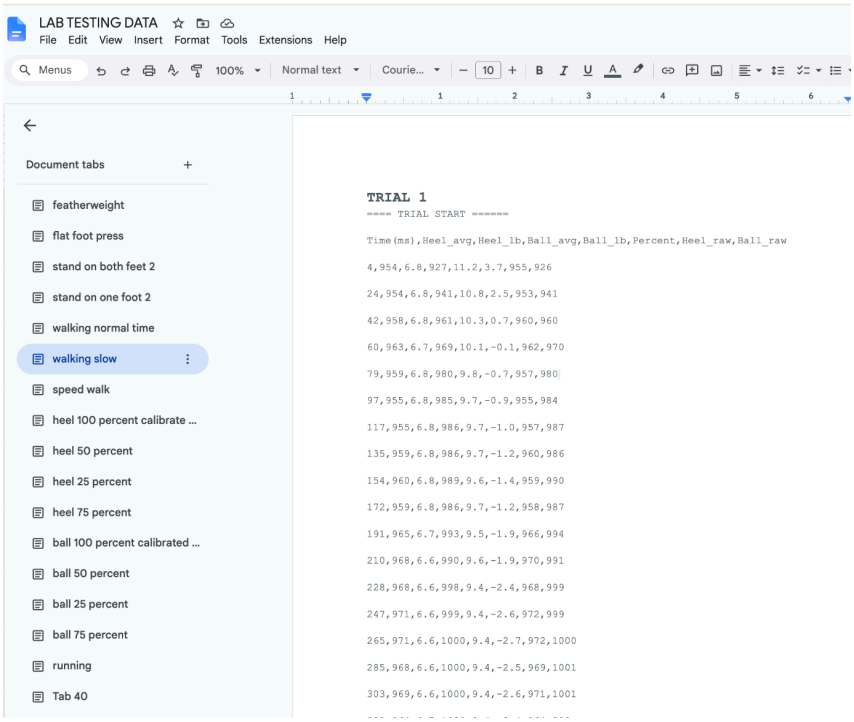
Heel 25% / 50% / 75% / 100% Calibration: These tests were different from the others in the way that the insole was taken out of the shoe. I used my heel to just press on the area where the heel sensor was for the known bodyweights above. There were three trials each, so 12 trials total.

Ball 25% / 50% / 75% / 100% Calibration: These tests were the same as the heel calibration but used on the ball sensor.



Data collection

As mentioned before, Ms. Costello and I would line up our times, and hit “go” at the same time. For the purpose of simplicity at the lab, I recorded all my data in google docs. Click here to see the full thing.



[Force Data]
NumberOfForcePlates=1
SampleRate=2000.000000
NumberOfSamples=8564

#Sample	FX1	FY1	FZ1	X1	Y1	Z1	MZ1
1	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
2	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
3	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
4	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
5	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
6	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
7	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
8	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
9	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
10	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
11	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
12	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
13	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
14	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
15	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
16	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
17	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
18	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
19	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
20	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
21	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
22	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
23	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
24	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
25	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
26	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
27	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
28	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
29	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
30	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
31	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
32	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
33	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
34	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
35	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
36	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
37	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
38	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
39	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
40	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
41	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
42	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
43	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
44	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
45	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
46	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00
47	-0.00	-0.00	-0.00	244.00	222.00	0.00	-0.00

Here is what the data looked like on the force plate (for the same walking slow trial). However, it wasn't recorded in seconds; it was recorded in sample numbers. The force plate takes 2000 samples per second. My device only reads about 53 times a second. So for every one reading of mine, the force plate took roughly 38.

After the lab, I organized all of the data cleanly to make it easy for analysis. I transferred all my data to a CSV file, and made a master folder on my desktop with all the data, called LAB_DATA. In that folder, it has 20 individual folders, 19 for each test, and one "other folder" for miscellaneous things. For each test folder, it has my csv file for that test (containing each trial), and another folder containing the three .forces files from the force plate. This master folder is on my github. On some tests, more than three trials were done (had to redo for some error), but for analysis only three were used.

Data Analysis

For data analysis, there were three "sets" I organized the analysis in:

Set one: Standing still (7 conditions, 21 trials)

This included 30%, 50%, 85%, featherweight, flat foot press, stand on both feet, and stand on one foot

Here was my plan in analyzing this data:

1. Read the force plate file → vertical force in newtons, 2000 times a second
2. Take the steady middle of the hold and average it.
3. Divide by 787.3 N (177lb, my bodyweight)→ the true percentage of my body weight
4. Read my device's CSV for the same trial
5. Take the median of its percent column over the same stretch → what my device said
6. Find the ratio between the two
7. Put both numbers in a clinical band (<10, 10–30, 30–50, 50–85, >85) and check whether they land in the same one

Step 7 matters because it is how I am testing the accuracy of the device. A doctor does not say a very specific number (ex: 47%), they say more generalizable numbers like 50%, or even terms like featherweight or flat foot press. Those are categories, not exact measurements, so to test my device I had to turn them into ranges: under 10%, 10–30%, 30–50%, 50–85%, and over 85%. An error of, say, 6 percentage points that stays inside the band is harmless, and an error of 6 percentage points that crosses one puts the patient in the wrong category.

I ran each trial's analysis in MATLAB. For the 21 standing trials, I loaded the raw .forces file for that trial, pulled out column 4 (the vertical force in newtons), trimmed one second

off the start and two off the end so I was only looking at the steady part of the hold, and averaged what was left. Dividing by 787.3N (my bodyweight) gave the true percentage. Then I opened up my device's CSV for the same trial, trimmed one second off the start and two off the end, then took the median body weight percentage. I used the median for my device to remove outliers.

Here is the code I used to look at one trial (flat foot press trial one), and the output:

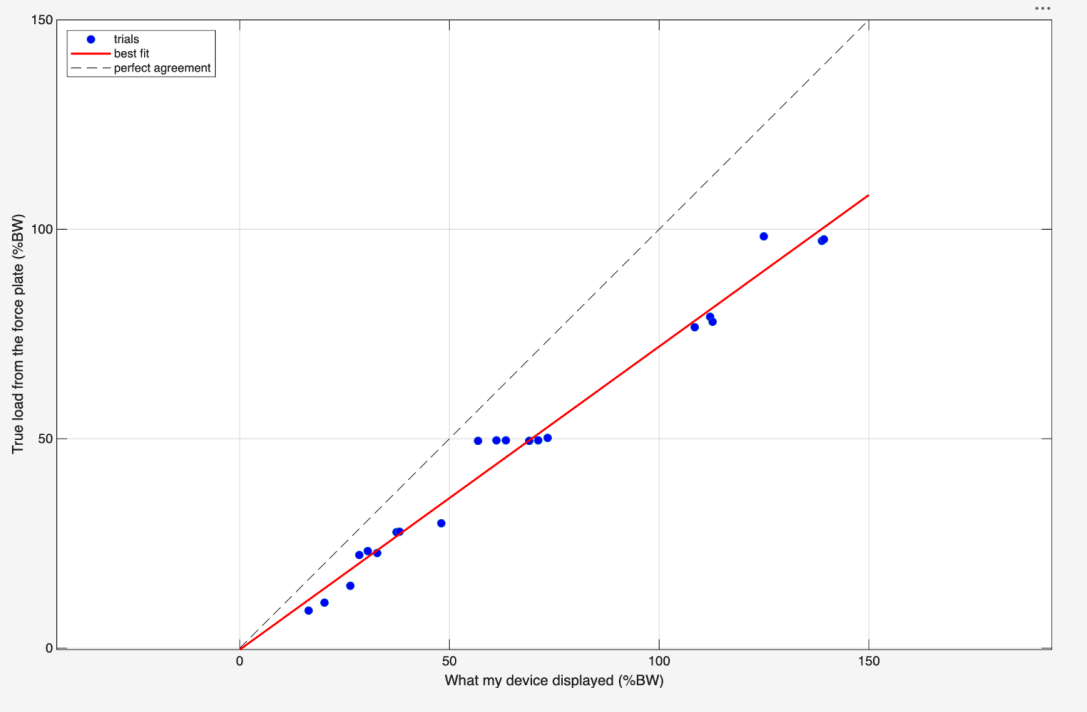
```
>> pf=fullfile(root,'flat foot press','lab_flat_foot_press','Michael_flat_foot_press_trial_1.forces');
fz=readmatrix(pf,'FileType','text','NumHeaderLines',5); fz=fz(:,4);
dev=readtable(fullfile(root,'flat foot press','my_data_flat_foot_press.csv')); p=dev.Percent(dev.Trial==1);
tp=mean(fz(2001:end-4000))/BW*100; dp=median(p(round(end*0.2):round(end*0.8)));
fprintf('flat foot press         T1   true %6.2f   device %7.2f   ratio %.3f\n',tp,dp,dp/tp);
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
flat foot press          T1   true  22.68   device   32.80   ratio 1.446
```

I put all the results into a csv file (standing_trials_results in github):

condition	trial	true_pct_BW	device_pct	ratio	true_band	device_band	band_match
featherweight	1	8.91	16.40	1.84	<10%	10-30%	NO
featherweight	2	10.87	20.20	1.86	10-30%	10-30%	YES
featherweight	3	14.90	26.40	1.77	10-30%	10-30%	YES
flat foot press	1	22.68	32.80	1.45	10-30%	30-50%	NO
flat foot press	2	23.18	30.50	1.32	10-30%	30-50%	NO
flat foot press	3	22.29	28.50	1.28	10-30%	10-30%	YES
30 percent	1	29.82	48.00	1.61	10-30%	30-50%	NO
30 percent	2	27.64	37.35	1.35	10-30%	30-50%	NO
30 percent	3	27.79	38.10	1.37	10-30%	30-50%	NO
50 percent	1	50.22	73.40	1.46	50-85%	50-85%	YES
50 percent	2	49.49	69.00	1.39	30-50%	50-85%	NO
50 percent	3	49.60	71.10	1.43	30-50%	50-85%	NO
stand on both feet	1	49.53	56.85	1.15	30-50%	50-85%	NO
stand on both feet	2	49.55	61.20	1.24	30-50%	50-85%	NO
stand on both feet	3	49.57	63.45	1.28	30-50%	50-85%	NO
85 percent	1	76.58	108.50	1.42	50-85%	>85%	NO
85 percent	2	79.11	112.10	1.42	50-85%	>85%	NO
85 percent	3	77.88	112.80	1.45	50-85%	>85%	NO
stand on test foot only	1	98.33	125.00	1.27	>85%	>85%	YES
stand on test foot only	2	97.29	138.80	1.43	>85%	>85%	YES
stand on test foot only	3	97.64	139.30	1.43	>85%	>85%	YES

Side note: For *stand on both feet*, both of my feet were on the force plate, so the plate measured my entire body weight rather than just the foot with the device in it. The plate read 99.06, 99.09 and 99.14 percent across the three trials, so I just divided those by 2 to correctly match them up with my data.

These results show that my device landed in the correct clinical band only 7 times out of 21, which is not good. However, every trial was reading high. This means this isn't likely noise. One consistent error in one direction means we can try to narrow the cause.



Above is the graph of what my device displayed against the force plate. If my device were accurate, plotting the true load against what it displayed would put every point on the diagonal. Instead every point sits below it. The best fit line has a slope of 0.72 and an R^2 of 0.98. Since the line is strongly linear but just shifted, this supported my theory that there was one consistent error.

The *stand on test foot only* test is the one that cracked it. Standing on one foot is the exact same thing my f command captures when it defines 100%. So my device should read about 100% for it, by definition. Instead it read 125.0%, 138.8% and 139.3% against a true load of 98.3, 97.3 and 97.6. My device was disagreeing with itself.

So the value stored as “100%” was wrong. The calibration samples themselves are gone - `f` averages the hold, saves one number, and throws the rest away. But *stand on test foot only* is the same action repeated later in the session, and that one was recorded. Solving the formula backwards on those three trials gave 48.40, 48.41 and 48.42 lb. For the device to have displayed the correct 97.8 percent, that number needed to be 59.47 lb. It was short by 11 pounds.

$$\text{percent} = (\text{heel_lb} + \text{ball_lb} - \text{baseline}) / (\text{fullWeight} - \text{baseline}) \times 100$$

The cause was obvious once I knew where to look. My f command averaged a five-second hold while I deliberately shifted my weight around, thinking several postures would give a better anchor. For part of that window less than my full weight was on the sensors, so the average came out low and that number is the denominator of every percentage the device reports.

To fix it, instead of averaging a five-second hold while I moved around, the code now takes the median of a steady hold, to get a more steady reading. I could have also just made a best fit line and a corrective constant, but I decided to fix the start of the problem, rather than working around it.

I could test this without going back to the lab, because my device recorded the force from each sensor, not just the final percentage. Every row of my CSV has `Heel_lb` and `Ball_lb`. The percentage was calculated from those two numbers using the anchor `f` captured - so I could take the forces my sensors actually measured and recompute every reading with the correct anchor instead.

Correcting the f value, the device correctly lands in 17/21 bands, with an average error of 3.30 percentage points, and it **and it never placed the load in a lower clinical band than the truth, which is the dangerous direction**

I also wanted to discuss the data from the force plate for clinical terms:

Term	What it's supposed to mean	What the plate measured
Featherweight	toe-touch, roughly 10–20% of body weight	8.9, 10.9, 14.9%
Flat foot press	around 25%	22.7, 23.2, 22.3%

Both landed around inside what the terms are supposed to mean, which is reassuring. But look at the featherweight spread: 8.9 to 14.9 percent across three tries, minutes apart, deliberately trying to do the same thing. This just goes to show the need for this project: even when I am trying hard and have a force plate in front of me, I cannot reproduce my own toe-touch

Set two: individual sensors (8 conditions, 24 trials)

To analyze the individual sensor data, I ran the same code in MATLAB with two changes. I converted the force plate reading into pounds instead of a percentage of body weight, and from my device's CSV I pulled the raw sensor reading rather than the pounds column. For the plate I trimmed the beginning and end and took the mean of the hold. For my device I took the median of the middle of the hold.

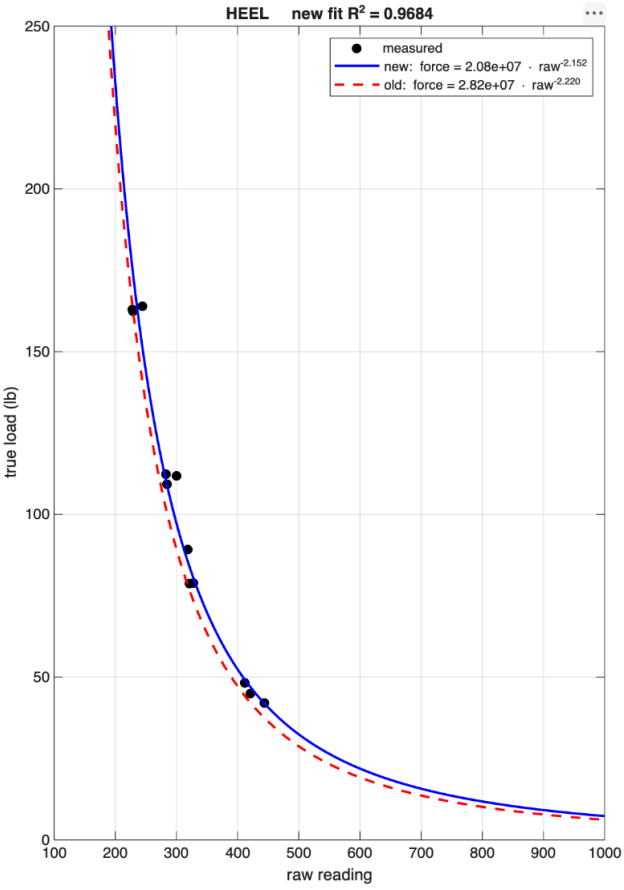
Here is an example input and output for ball sensor full bodyweight trial 1:

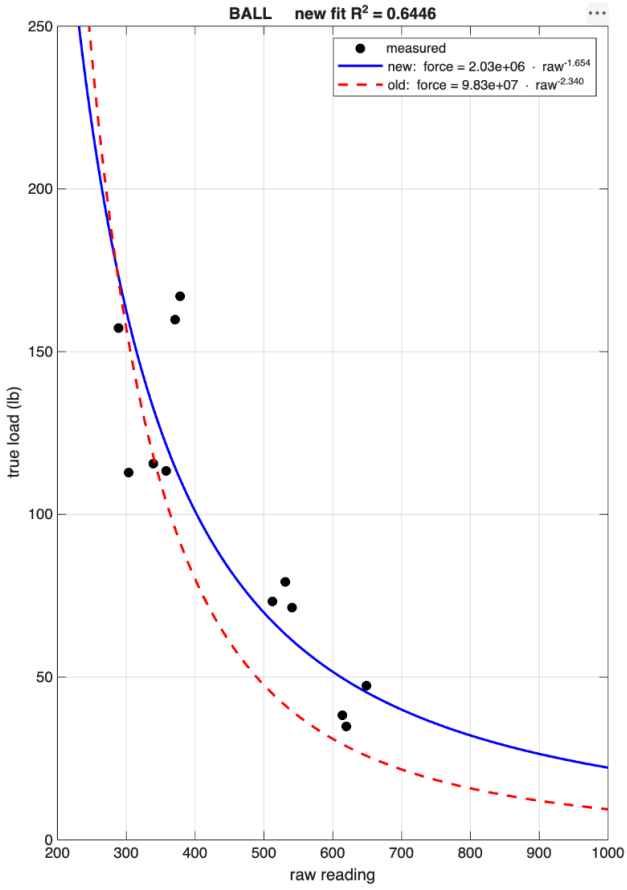
```
>> pf=fullfile(root,'ball sensor full body weight','lab_ball_sensor_full_body_weight','Michael_ball_sensor_full_body_weight_trial_1.forces')
fz=readmatrix(pf,'FileType','text','NumHeaderLines',5); fz=fz(:,4);
dev=readtable(fullfile(root,'ball sensor full body weight','my_data_ball_sensor_full_body_weight.csv')); r=dev.Ball_avg(dev.Trial==1);
tl=mean(fz(2001:end-4000))/N2LB; raw=median(r(round(end*0.2):round(end*0.8)));
fprintf('ball sensor full body weight   T1   raw %4.0f   true %6.1f lb\n',raw,tl);
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
ball sensor full body weight   T1   raw  371   true  159.8 lb
```

sensor	level	trial	raw_reading	true_lb
heel	25%	1	421	45.0
heel	25%	2	412	48.2
heel	25%	3	444	42.1
heel	50%	1	328	78.9
heel	50%	2	318	89.2
heel	50%	3	321	78.8
heel	75%	1	283	112.3
heel	75%	2	285	109.3
heel	75%	3	300	111.9
heel	100%	1	244	164.0
heel	100%	2	229	162.4
heel	100%	3	228	162.9
ball	25%	1	649	47.3
ball	25%	2	620	34.8
ball	25%	3	614	38.2
ball	50%	1	531	79.3
ball	50%	2	541	71.3
ball	50%	3	512	73.3
ball	75%	1	339	115.6
ball	75%	2	358	113.3
ball	75%	3	303	112.8
ball	100%	1	371	159.8
ball	100%	2	289	157.2
ball	100%	3	378	167.0

Here are the results (sensor_calibration_trials.csv)

From this I used MATLAB to create new calibration graphs and compared them to my old ones:





The new heel curve is a verification. It was basically the same as the old curve I made myself. There was two calibrations, different equipment, months apart, and produced nearly the same curve

The ball sensor isn't as good as the heel, but we also knew this from before. R^2 of 0.645 means a third of the variation in load isn't explained by the raw reading. The readings go backwards between 75% and 100%, and the same press at 100% gave 371, 289, and 378.

However, what surprised me the most was I expected to leave this lab with two new calibration curves, but instead I decided to just stick with my old ones.

Using MATLAB, I backsolved to see how well these new curves would have been at the lab.

```
=== OLD curves ===
  baseline 19.66 lb   fullWeight 59.84 lb
  bands 14/15   under-banded 0   MAE 3.53 pts
   flat foot press          true  22.68  23.18  22.29  |  dev   20.43   18.71   17.23
   30 percent               true  29.82  27.64  27.79  |  dev   32.01   23.64   24.44
   50 percent               true  50.22  49.49  49.60  |  dev   51.37   47.95   49.45
   stand on both feet       true  49.53  49.55  49.57  |  dev   40.66   41.96   43.81
   85 percent               true  76.58  79.11  77.88  |  dev   78.08   80.81   81.31

=== NEW curves ===
  baseline 35.27 lb   fullWeight 82.48 lb
  bands 14/15   under-banded 0   MAE 3.62 pts
   flat foot press          true  22.68  23.18  22.29  |  dev   19.41   17.64   15.64
   30 percent               true  29.82  27.64  27.79  |  dev   33.45   24.58   25.67
   50 percent               true  50.22  49.49  49.60  |  dev   51.70   48.08   49.48
   stand on both feet       true  49.53  49.55  49.57  |  dev   41.33   41.61   42.81
   85 percent               true  76.58  79.11  77.88  |  dev   77.22   79.91   80.51

OLD 14/15 (0 under)   NEW 14/15 (0 under)
>>
```

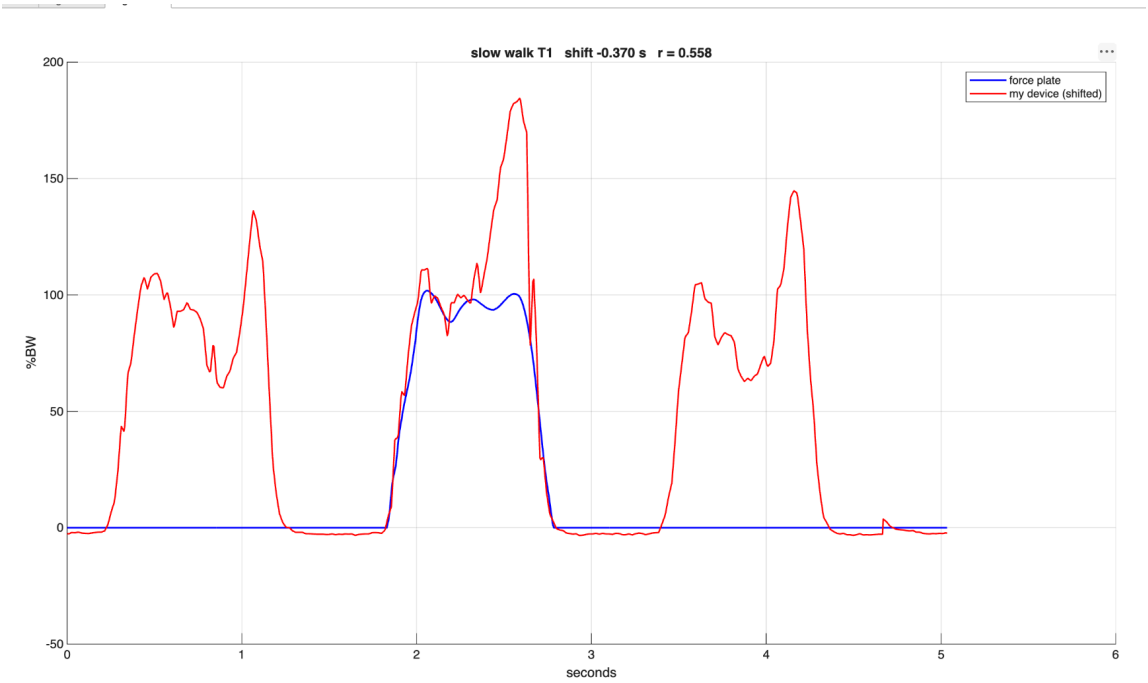
I solved for both constants using the two conditions whose true answers I know independently (featherweight and standing on the test foot only) and scored the other 15 trials under each curve set. It was a tie: 14 of 15 bands and zero under-bands either way, with mean error 3.53 points on the old curves and 3.62 on the new. The curve choice made no measurable difference to the output, which is what you'd expect from a device that reports a ratio (a curve wrong by a consistent factor cancels out of it).

So I changed nothing. The heel curve was confirmed, the ball curve couldn't be improved with the data I had, and switching would have invalidated the 17 of 21 results found in the Standing Still part for no measurable gain.

Set three: moving (4 conditions, 12 trials)

The purpose of this set is to test the device in the activity it actually exists for - walking

Because Ms. Costello and I both started collecting data after one of us said “go,” I knew that it would not perfectly line up. We were off by a few tenths of a second. Furthermore, the forceplate only recorded one stride, while my device recorded multiple across the room. So I used MATLAB to align the force plate’s curve with mine - it aligned with the closest jump.



This is trial one of slow walking, and it actually provides some interesting results. For one, you can see the triple hump (or triple peak) pattern. This was different from the double hump we expected, and a middle peak probably showed up because I was walking slowly. With less vertical momentum the two main humps flatten out, leaving small balance corrections visible as their own bump.. You can see this pattern in both my device and the force plate. The overlaid walking step actually aligns pretty well in the beginning, but is inaccurate at the end. This is most definitely because the first peak is due to the heel stepping down, and the last peak is due to pushing off with the ball of the foot. The accurate heel sensor and inaccurate ball sensor shows the divergence between the two traces in the second half of the step in the graph.

I did the same for the rest of the trials. See picture below. Red is my device, blue is the force plate:



For slow walking trial two, the triple hump pattern showed again on the force plate, but there was a clear double hump on my device. Both of the humps were relatively close to each other, even though the middle wasn't. In trial three, my device completely spiked above 200%, which was probably some sort of sensing error.

Walking normal had precise results across the three trials. The first hump was pretty close and the second hump was greater.

For speed walking trials two and three, the middle of the graph was almost exactly overlaid on top of each other.

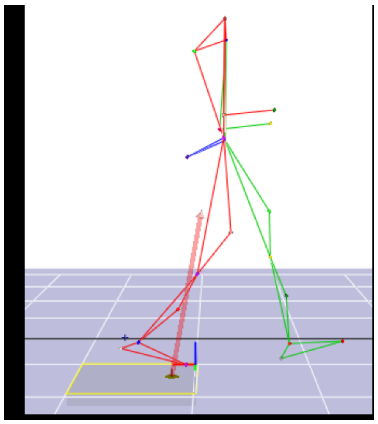
Running was decently close, but again, my device peaked a lot. Honestly, running was just done at the very end as a sort of "fun" trial. There's nobody in rehab who is going to be running with this device on.

Furthermore, in all these trials you can see both my device and the force plate went above 100% body weight in most of the peaks. This is reasonable because when you walk you aren't just holding your weight up, you're also accelerating your body up and down, and the ground has to supply that extra force on top of your body weight.

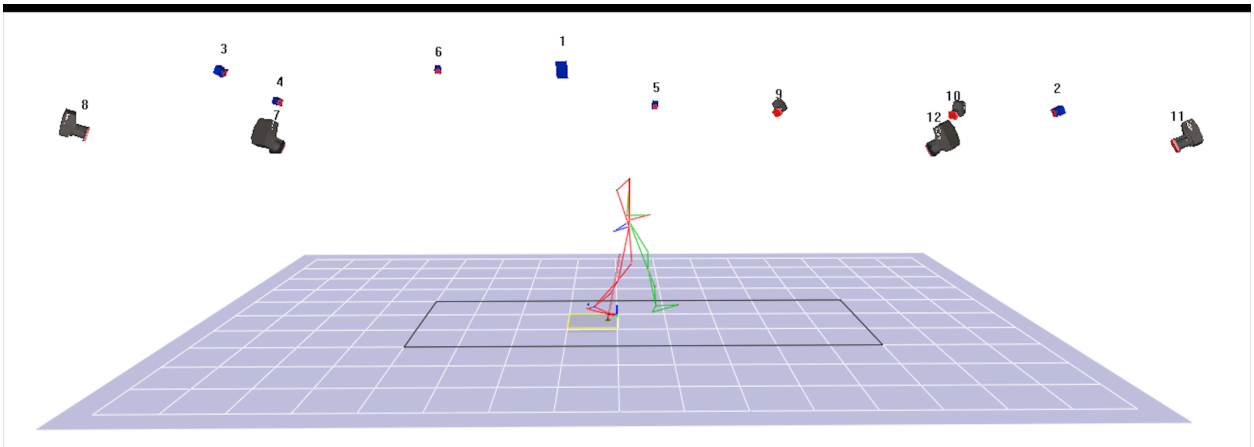
I also used MATLAB to measure the time shift needed to align the two recordings, how closely the curves correlated within the stance phase, the peak each instrument recorded, the ratio between those peaks, and how my device compared to the plate across the first, middle and last third of each step.

```
condition          tr  shift    r_st   plate%  dev%  ratio  early  mid   late  satN
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
slow walk           1  -0.37   0.821    101.8  184.5   1.81   1.09  1.12  1.59    26
slow walk           2  -0.40   0.762    105.7  124.4   1.18   1.11  0.71  1.10    12
slow walk           3  -0.31   0.603    103.5  350.2   3.38   1.06  1.14  2.17    19
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
walking normal      1  -0.49   0.852    115.4  189.0   1.64   1.02  0.81  1.24    19
walking normal      2   1.07   0.736    119.0  150.8   1.27   1.06  0.88  1.09     9
walking normal      3   0.80   0.694    120.3  155.3   1.29   1.10  0.89  1.13    10
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
speed walk          1   1.51   0.912    153.6  251.1   1.63   1.53  0.27  0.92    11
speed walk          2  -1.12   0.939    151.8  209.6   1.38   0.99  1.00  1.33     8
speed walk          3  -1.27   0.961    146.8  199.9   1.36   1.05  0.85  1.27     6
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The
original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.
run                 1  -1.59   0.411    187.3  382.4   2.04   1.78  0.68  1.14   110
run                 2  -1.83   0.563    187.2  298.8   1.60   1.49  0.75  1.05    87
run                 3  -1.75   0.529    187.2  360.3   1.93   1.62  0.67  1.04    97
>>
```

These numbers line up with what the graphs show. Every peak ratio was above 1.0, so my device never under-reads at the top of a step, but splitting each stance into thirds shows the error isn't even. My device averaged about 1.24× the plate early, 0.81× in the middle, and 1.26× late, with the middle lowest in nine of the twelve trials. That is the high-low-high shape you can see in the panels. It overshoots at heel strike, dips through midstance, then overshoots again at push-off.



Here are some fun images from the gate-analysis



In conclusion:

The lab visit was extremely beneficial. In total we ran 19 tests and 57 trials. From them I learned that my f calibration command was storing a value too low, which was making every reading high; that my calibration curves were not the problem; and that my two sensors miss the load during midstance, when it sits in the arch between them. Fixing the f command moved my device from landing in the correct clinical band 7 times out of 21 to 17 out of 21, with the average error dropping from 18.8 percentage points to 3.30, and it never placed the load in a lower band than the truth. Standing still now works, and moving kind of follows the real pattern. I think one of the main things holding this device back is the inaccurate ball sensor. If there was another sensor like the heel sensor, then it would be amazing. This is the tradeoff for getting cheap FSRs (they vary unit to unit, and the heel one happened to be good while the ball one wasn't). But the good thing is now we know that my at home calibration works (it worked for the heel sensor), so I could get another sensor and re-calibrate that for the ball.

For the next prototyping phase, I will compact the device. After that, my next steps are to get another sensor and calibrate that.

Overall, this was an incredible experience going to the lab. HUGE thanks to both Dr. Jenkyn and Ms. Costello for helping out (and my dad for driving)!

